

# Vulnerability Assessment Report

1<sup>st</sup> January 20XX

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## System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

## Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

## Purpose

*The purpose of this vulnerability assessment is to identify security risks related to the company's publicly accessible database server. The database server is critical to business operations because remote employees rely on it to access customer and sales information. If the server is compromised or becomes unavailable, it could negatively impact daily operations, customer trust, and the company's reputation. This assessment will help identify vulnerabilities and recommend security controls such as multi-factor authentication, access controls, and backup systems to better protect sensitive business data.*

## Risk Assessment

| Threat source   | Threat event                                  | Likelihood | Severity | Risk |
|-----------------|-----------------------------------------------|------------|----------|------|
| E.g. Competitor | Obtain sensitive information via exfiltration | 1          | 3        | 3    |

|                        |                                                             |   |   |   |
|------------------------|-------------------------------------------------------------|---|---|---|
| <i>Outsider/Hacker</i> | Perform reconnaissance and surveillance of the organization | 3 | 2 | 6 |
| <i>Outsider/Hacker</i> | Obtain sensitive information via exfiltration               | 3 | 3 | 9 |
| <i>Outsider/Hacker</i> | Conduct Denial of Service (DoS) attacks                     | 3 | 3 | 9 |
| <i>Outsider/Hacker</i> | Alter/Delete critical information                           | 3 | 3 | 9 |

## Approach

The selected threat events were chosen because the database server is publicly exposed to the internet, which increases the likelihood of external attacks. Reconnaissance and surveillance activities are significant because attackers often begin by scanning and analyzing exposed systems before attempting further attacks. If vulnerabilities are discovered, attackers may obtain sensitive information through exfiltration, disrupt operations through Denial of Service (DoS) attacks, or alter and delete critical business data. These risks could negatively affect business operations, customer trust, and the overall security of the organization.

## Remediation Strategy

To reduce the identified risks, the organization should implement stronger access controls and limit public exposure of the database server. Access to the database should only be allowed for authorized employees using multi-factor authentication, strong passwords, and role-based access controls based on job responsibilities. The company should also use firewall rules, IP allow-listing, or VPN access to prevent unauthorized external connections and reduce reconnaissance attempts. In addition, regular backups, monitoring, and auditing systems should be implemented to improve recovery capabilities and detect suspicious activity.